

Linear programmes: a primer sheet

The paper version of the optional LP primer of the self-study website (before step 10). It needs a pencil, a ruler and a calculator. The prices, contents and caps are invented for the exercise; they are not market data. Amounts are kilograms and contents are mass fractions; step 10 uses moles of atoms and atom fractions, and the recipe is the same.

1. Three parts

A melt shop must make 1 kg of a copper-nickel alloy with exactly 0.40 kg of nickel in it, as cheaply as possible, from these lots:

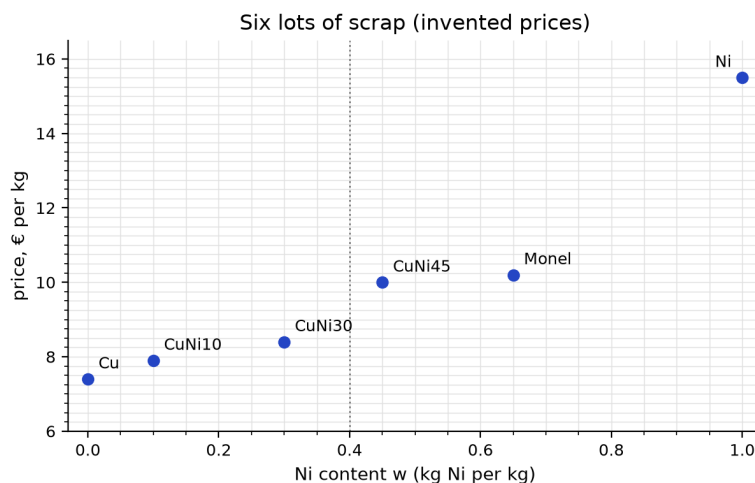
Lot	Ni content w (kg Ni per kg)	Price (€ per kg)
Copper scrap (Cu)	0.00	7.40
CuNi10 scrap	0.10	7.90
CuNi30 scrap	0.30	8.40
Constantan scrap (CuNi45)	0.45	10.00
Monel offcuts	0.65	10.20
Nickel cathode (Ni)	1.00	15.50

Every linear programme (LP) has three parts:

- **Choose** the amounts $f_j \geq 0$ of each lot, in kg (the variables);
- **to make** the total cost $\sum_j f_j c_j$ as small as possible (the objective);
- **while keeping** the rules $\sum_j f_j = 1$ and $\sum_j f_j w_j = 0.40$ (the constraints).

Books, papers and software write it in one line: minimise $\sum_j f_j c_j$ **subject to** $\sum_j f_j = 1$, $\sum_j f_j w_j = 0.40$, $f_j \geq 0$. "Subject to" means "while keeping these rules". A choice that keeps every rule is *feasible*; any feasible choice with the lowest total is *optimal*. *Linear* means that the total and every rule are sums of amounts times fixed numbers.

2. Dots and chords



1. Which lots have less nickel than 0.40, and which more? Use the lever rule on CuNi10 and CuNi30: why can this pair not make 0.40?

2. With a ruler, join Cu and Ni cathode, Cu and Monel, CuNi30 and Monel. Read each line's height at 0.40, then check it with the lever rule.
3. Which of the nine possible pairs is cheapest? How many lots does it use?

3. How a computer searches

Start from Cu and Ni cathode. Draw the line through the two used lots. The lot furthest below the line (measured straight down, in €) comes in; the used lot on the same side of 0.40 goes out. Repeat until no lot lies below the line.

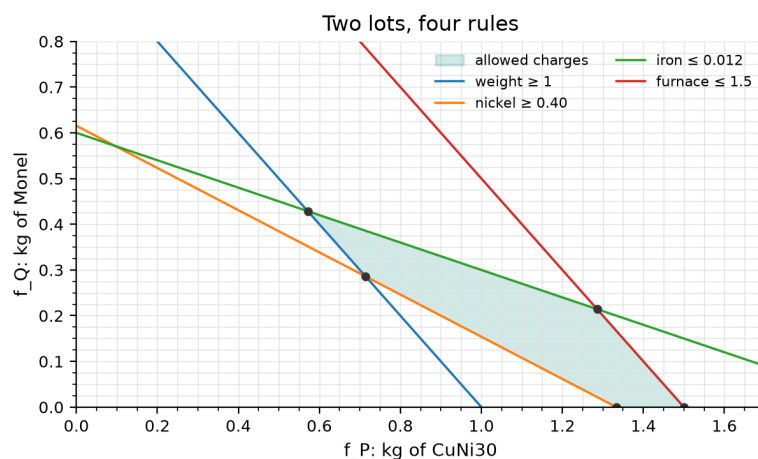
Round	Used pair	Cost (€ per kg)	Lot furthest below the line	Swap
1	Cu + Ni cathode			
2				
3				

4. Prices and a floor

1. Extend the final line to $w = 0$ and $w = 1$. What is 1 kg of copper worth in this charge, and 1 kg of nickel? Why is the nickel cathode not used?
2. Nudge the target from 0.40 to 0.41. Predict the new cost from the line's slope. For which targets does this price hold?
3. Explain in two sentences why no charge at 0.40 can cost less than the line's height there.
4. A dealer offers swarf at $w = 0.50$ for €9.20. Is it worth buying? Answer from the line, without solving again.

5. The textbook picture

Now only two lots, CuNi30 (amount f_P) and Monel (amount f_Q), with "at least" and "at most" rules: the charge weighs at least 1 kg; it holds at least 0.40 kg nickel ($0.30f_P + 0.65f_Q \geq 0.40$); at most 0.012 kg iron ($0.006f_P + 0.020f_Q \leq 0.012$); the furnace holds at most 1.5 kg; no amount is negative.



The five corners of the allowed polygon:

Corner (f_P, f_Q), kg	Tight rules	Cost (€)
(0.714, 0.286)	weight, nickel	
(0.571, 0.429)	weight, iron	
(1.286, 0.214)	iron, furnace	
(1.500, 0)	furnace, no Monel	
(1.333, 0)	nickel, no Monel	

1. Fill in the cost $8.40f_P + 10.20f_Q$ at each corner.
2. Draw the cost line $8.40f_P + 10.20f_Q = 9.00$. Slide it, parallel, to lower cost until it is about to leave the polygon. Which corner does it touch last?
3. Which rules have room to spare at that corner? What is their price? (A rule's price is how much the best cost changes when the number on its right-hand side moves by one unit.)
4. The iron cap becomes 0.005 kg. What happens?

6. Same problem, new names

This sheet	Step 10 onwards
a lot of scrap	a state: one phase model at one composition
nickel content w	B atom fraction x
price c , € per kg	molar Gibbs energy g , J/mol atoms
weight and nickel rules	all atoms and all B atoms are kept
worth of 1 kg Cu and 1 kg Ni	μ_A and μ_B
price of nickel content	$\Delta\mu = \mu_B - \mu_A$
a new offer below the line	a missing state found by pricing

One difference: costs add because lots are bought and melted separately. A mixture of menu dots adds energies the same way, as a real sample of separate regions (boundary energy ignored), so the truth is at or below it: a ceiling. It need not be the truth, because states between the dots are missing: if the atoms of two dots of one phase model form one phase at the average composition, its energy lies on that phase's curve, which can lie below the chord.

Answers (after your attempt)

2.1 Cu, CuNi10 and CuNi30 lie below 0.40; Constantan, Monel and Ni cathode above. For CuNi10 and CuNi30 the lever rule asks for 1.5 kg CuNi30 and -0.5 kg CuNi10: a negative amount cannot be bought.

2.2 Cu + Ni cathode: 0.600 + 0.400 kg, €10.64. Cu + Monel: 0.385 + 0.615 kg, €9.12. CuNi30 + Monel: 0.714 + 0.286 kg, €8.91.

2.3 CuNi30 + Monel, €8.914 per kg. Two lots, and no more are needed: a three-lot charge at 0.40 is a mixture of two-lot charges, each at 0.40, and the cheaper of those costs no more than the mixture.

3 Round 1: Cu + Ni cathode, €10.64; Monel lies 2.47 below; Monel in, Ni cathode out. Round 2: Cu + Monel, €9.12; CuNi30 lies 0.29 below; CuNi30 in, Cu out. Round 3: CuNi30 + Monel, €8.91; no lot below the line: done.

- 4.1** The line's ends: €6.86 at $w = 0$ and €12.00 at $w = 1$. In this charge 1 kg Cu is worth €6.86 and 1 kg Ni €12.00. Ni cathode costs €15.50, more than it is worth here.
- 4.2** The slope is 5.14 € per unit of content, so the cost rises by $5.143 \times 0.01 = 0.051$ € to €8.966. The price holds while CuNi30 and Monel stay in use: targets between 0.30 and 0.65. At 0.30 and 0.65 themselves the target sits on a used lot, and the rates on either side differ.
- 4.3** Every lot costs at least the line's height at its own content. A charge is a mixture of lots whose amounts add to 1 kg and whose nickel adds to 0.40 kg, so it costs at least the line's height at 0.40, €8.914.
- 4.4** The line's height at 0.50 is €9.43; the swarf lies €0.23 below it, so it is worth buying. Half CuNi30 and half swarf cost €8.80.
- 5.1** €8.91, €9.17, €12.99, €12.60, €11.20.
- 5.2** The corner (0.714, 0.286), €8.914: the same charge as in part 2.
- 5.3** The iron and furnace rules; their price is zero. The weight rule costs €6.86 per kg, the nickel rule €5.14 per kg of nickel: the line's two numbers. More nickel in the same weight swaps copper (€6.86) for nickel (€12.00), so $12.00 - 6.86 = 5.14$.
- 5.4** No charge keeps every rule: the problem is infeasible, and a solver says so instead of giving an answer.